National
Qualifications
2021 ASSESSMENT RESOURCE

**X847/76/12**

# Mathematics
# Paper 2

Duration — 1 hour 30 minutes

**Total marks — 65**

**SECTION 1 — 52 marks**

Attempt ALL questions.

**SECTION 2 — 13 marks**

Attempt EITHER Part A OR Part B.

**You may use a calculator.**

To earn full marks you must show your working in your answers.

State the units for your answer where appropriate.

You will not earn marks for answers obtained by readings from scale drawings.

Write your answers clearly in the spaces provided in the answer booklet. The size of the space provided for an answer is not an indication of how much to write. You do not need to use all the space.

Additional space for answers is provided at the end of the answer booklet. If you use this space you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

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## FORMULAE LIST

### Circle

The equation $x^2 + y^2 + 2gx + 2fy + c = 0$ represents a circle centre $(-g, -f)$ and radius $\sqrt{g^2 + f^2 - c}$.

The equation $(x-a)^2 + (y-b)^2 = r^2$ represents a circle centre $(a, b)$ and radius $r$.

**Scalar product** $\mathbf{a.b} = |\mathbf{a}||\mathbf{b}|\cos\theta$, where $\theta$ is the angle between $\mathbf{a}$ and $\mathbf{b}$

or $\mathbf{a.b} = a_1b_1 + a_2b_2 + a_3b_3$ where $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$.

**Trigonometric formulae**

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= 2\cos^2 A - 1$$

$$= 1 - 2\sin^2 A$$

**Table of standard derivatives**

| $f(x)$ | $f'(x)$ |
|---|---|
| $\sin ax$ | $a\cos ax$ |
| $\cos ax$ | $-a\sin ax$ |

**Table of standard integrals**

| $f(x)$ | $\int f(x)dx$ |
|---|---|
| $\sin ax$ | $-\frac{1}{a}\cos ax + c$ |
| $\cos ax$ | $\frac{1}{a}\sin ax + c$ |

MARKS

## SECTION 1 — 52 marks

**Attempt ALL questions**

**1.** Determine the equation of the tangent to the curve $y = 2x^3 - 8x^2 + 14$ at the point where $x = 3$. **4**

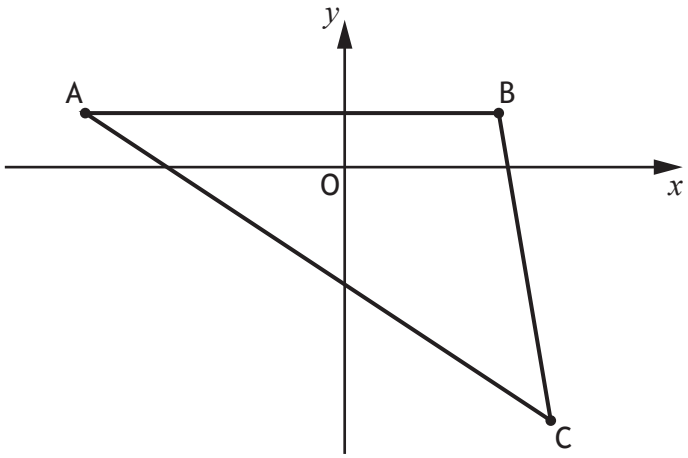
**2.** Find $\int \frac{6}{(x+5)^{\frac{3}{2}}}\,dx,\ x > -5$. **3**

**3.** Given $h(t) = \sin\left(2t + \frac{\pi}{6}\right)$, determine the rate of change of $h$ when $t = 10$. **3**

**[Turn over**

MARKS

**4.** Triangle ABC has vertices A (–5, 1), B (3, 1) and C (4, –5).




(a) The line $L_1$ is the altitude through B.
Find the equation of $L_1$. **3**

(b) The line $L_2$ is the perpendicular bisector of AB.
Find the equation of $L_2$. **3**

(c) Determine the coordinates of the point of intersection of $L_1$ and $L_2$. **1**

**5.** (a) Express $3\cos t° + 5\sin t°$ in the form $k\sin(t+a)°$, $k > 0$, $0 < a < 360$. **4**

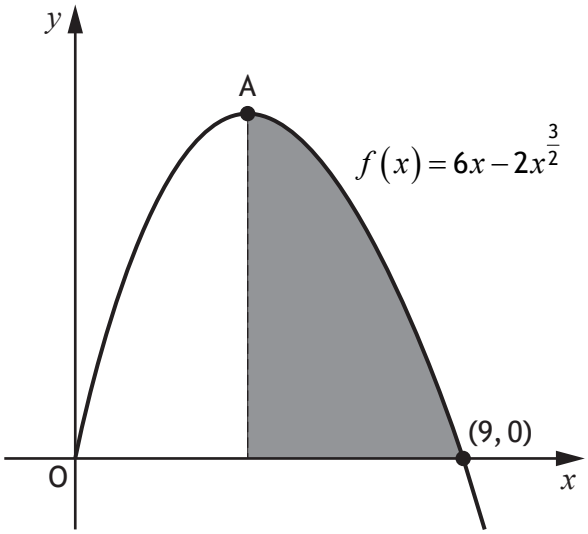
(b) A function, $f$, is defined by $f(t) = 3\cos t° + 5\sin t°$, $0 \le t < 360$.

(i) State the minimum value of $f(t)$. **1**

(ii) Determine the value of $t$ where this minimum occurs. **1**

MARKS

**6.** The graph of the function $f(x) = 6x - 2x^{\frac{3}{2}}$, $x \geq 0$ is shown.
The point A is a stationary point of $f(x)$.



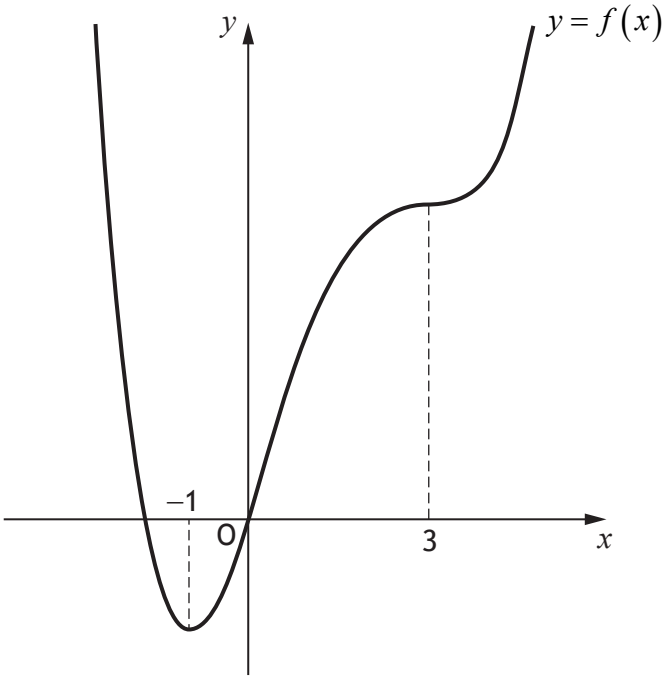

(a) Determine the $x$-coordinate of the stationary point A. **3**

(b) Hence calculate the shaded area. **4**

**[Turn over**

MARKS

**7.** The diagram shows the graph of $y = f(x)$, which has stationary points at $x = -1$ and $x = 3$.




On the diagram in your answer booklet, sketch a possible graph of $y = f'(x)$. **3**

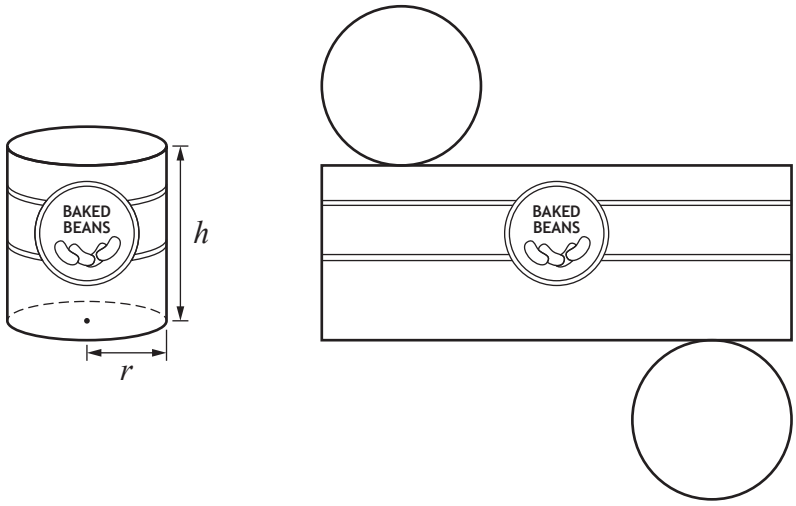
**8.** Solve the equation $2\sin(3x - 60)° + 1 = 0$, $0 \leq x < 180$. **4**

MARKS

**9.** A cylindrical tin of baked beans has a volume of 450 $cm^3$.

The radius of the tin is $r$ cm and its height is $h$ cm.

A net of the tin is shown in the diagram.




(a) Show that the surface area of the tin, $A$ square centimetres, is given by

$$A(r) = 2\pi r^2 + \frac{900}{r}.$$ **3**

(b) Determine the radius that will minimise the surface area. **6**

**10.** (a) Show that $2\tan x\cos^2 x = \sin 2x$, where $-\frac{\pi}{2} < x < \frac{\pi}{2}$. **2**

(b) Given that

- $\frac{dy}{dx} = 6\tan x\cos^2 x$, and
- $y = 3$ when $x = 0$,

express $y$ in terms of $x$. **4**

**[END OF SECTION 1]**

**[Turn over**

MARKS

## SECTION 2 — 13 marks

**Attempt EITHER Part A OR Part B**

**Part A**

**11.** (a) Given A (3, 1, 8), B (–2, 5, 1) and C (7, –6 , 3),

express $\overrightarrow{AB}$ and $\overrightarrow{AC}$ in component form. **2**

(b) Hence calculate the size of angle BAC. **4**

**12.** A sequence of real numbers is such that

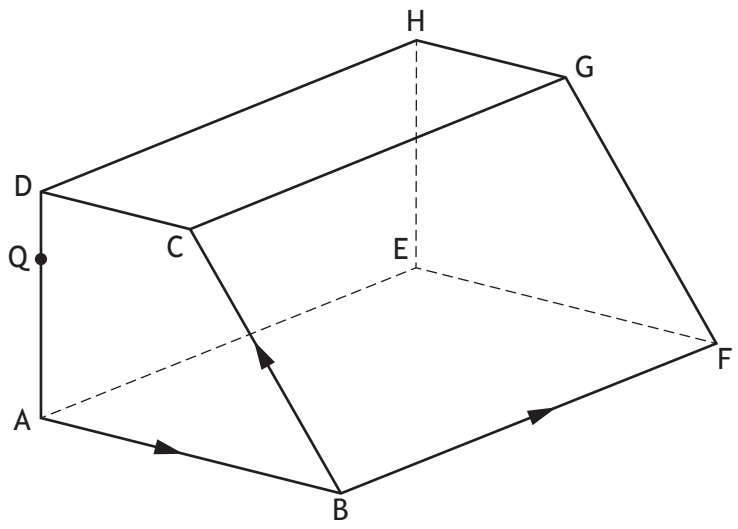
- the terms of the sequence satisfy the recurrence relation $u_{n+1} = 9u_n - 440$
- $u_{n+1} > u_n$ for all values of $n$.

The difference between two particular terms, $u_{k+1}$ and $u_k$, is 1000.

Determine, algebraically, the value of $u_k$. **3**

MARKS

**13.** ABCD,EFGH is a prism.




- $\overrightarrow{AB} = \begin{pmatrix} 8 \\ -4 \\ 6 \end{pmatrix}$, $\overrightarrow{BC} = \begin{pmatrix} -7 \\ 5 \\ 3 \end{pmatrix}$ and $\overrightarrow{BF} = \begin{pmatrix} 7 \\ 11 \\ -2 \end{pmatrix}$.
- $\overrightarrow{AB} = 2\overrightarrow{DC}$.

(a) Express $\overrightarrow{CF}$ in component form. **1**

(b) Hence, or otherwise, express $\overrightarrow{DF}$ in component form. **1**

(c) The point Q lies on the line AD.

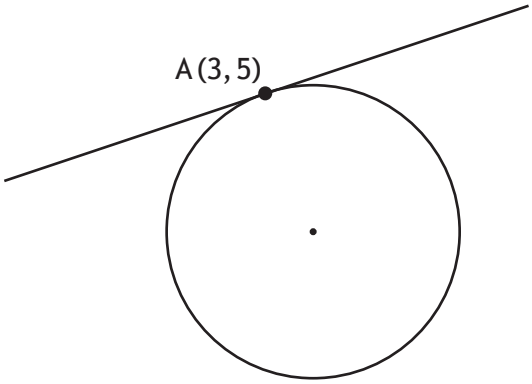
Given that $\overrightarrow{QF} = \begin{pmatrix} 17 \\ 5 \\ 0 \end{pmatrix}$, find $\overrightarrow{QD}$. **2**

**[Turn over**

MARKS

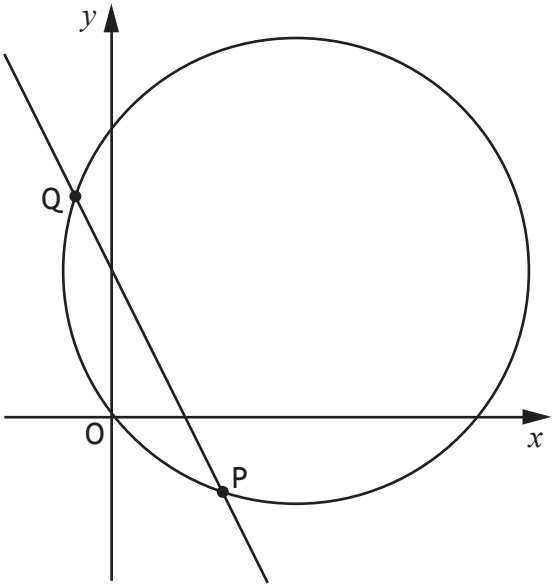
**Part B**

**14.** The point A (3, 5) lies on the circle with equation $x^2 + y^2 - 10x + 2y - 14 = 0$.




Find the equation of the tangent to the circle at A. **4**

**15.** The line $y = 4 - 2x$ intersects the circle $x^2 + y^2 - 10x - 8y + 1 = 0$ at the points P and Q.

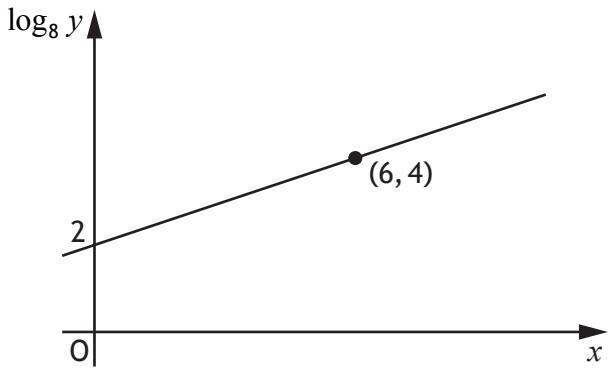



Find the coordinates of the points of intersection. **4**

MARKS

**16.** Two variables, $x$ and $y$, are connected by the equation $y = ab^x$.

The graph of $\log_8 y$ against $x$ is a straight line as shown.




Find the values of $a$ and $b$. **5**

**[END OF SECTION 2]**

**[END OF QUESTION PAPER]**